

## Digital – to – Analog converter. (DAC)

### ✪ Design a weighted-resistor D/A converter and explain how it works.

The circuit of Fig. 10.1 can be used for converting the digital input to analog output which operates according to the above principle. This circuit is referred to as a **weighted-resistor D/A converter** since the resistance values are weighted in accordance with the binary weights.

In the circuit Fig. 10.1, the digital inputs (1 or 0) operate the switches. A switch is thrown to position 1 or 0 for a digital input corresponding to that bit being 1 or 0,

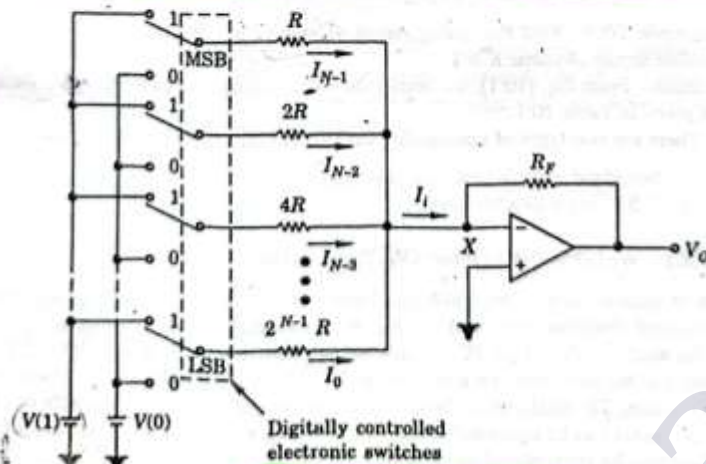


Fig. 10.1 Weighted-resistor D/A converter.

respectively. The voltage applied to a resistor is  $V(1)$  if the switch connected to it is in position 1 and  $V(0)$  if it is in position 0. The current,  $I_i$  is given by

$$I_i = I_{N-1} + I_{N-2} + I_{N-3} + \dots + I_2 + I_1 + I_0 \quad (10.2)$$

where

$$\left. \begin{aligned} I_{N-1} &= V_{N-1} / R \\ I_{N-2} &= V_{N-2} / 2R \\ I_{N-3} &= V_{N-3} / 2^2 R \\ &\vdots \\ I_0 &= V_0 / 2^{N-1} R \end{aligned} \right\} \text{where } V_n = V(1) \text{ if } b_n = 1 \\ = V(0) \text{ if } b_n = 0 \quad (10.3)$$

For straight binary input,  $V(0) = 0$  and  $V(1) = -V_R$ , and the output voltage  $V_O$  is given by

$$V_O = -(-V_R) \left( \frac{R_F}{R} b_{N-1} + \frac{R_F}{2R} b_{N-2} + \frac{R_F}{2^2 R} b_{N-3} + \dots + \frac{R_F}{2^{N-1} R} b_0 \right) \quad (10.4)$$

which is of the same form as Eq. (10.1)

with

$$K = \frac{R_F}{2^{N-1} R} \cdot V_R \quad (10.5)$$

The output swings in only one direction and therefore is *unipolar*. If it is required to convert digital data in bipolar format such as in sign-magnitude, 1's complement or 2's complement format, then  $V(0) \neq 0$ . In such cases,  $V(0)$  is used to *offset* the output swing.

With  $V(1)$  and  $V(0)$  as the voltages applied to the resistor network for 1 and 0, respectively, the output voltage  $V_O$  of Fig. 10.1 is given by

$$V_O = \frac{R_F}{2^{N-1} R} (2^{N-1} V_{N-1} + 2^{N-2} V_{N-2} + \dots + 2^1 V_1 + 2^0 V_0) \quad (10.6)$$

- ✪ **Find the analog output voltage of a 3-bit D/A converter for all possible inputs. Assume K=1.**  
Answer is partially matched with the weighted-resistor D/A converter where we deduce a general equation. Following this equation we make K=1 and N = 3 then we find output voltage for each combination.

For a 3-bit DAC, there are  $2^3 = 8$  possible input combinations. Let's represent the digital input as  $b_2, b_1, b_0$ , where  $b_2$  is the most significant bit and  $b_0$  is the least significant bit.

The analog output voltage can be calculated using the following formula:

$$V_{out} = K \times (b_2 \times 2^2 + b_1 \times 2^1 + b_0 \times 2^0)$$

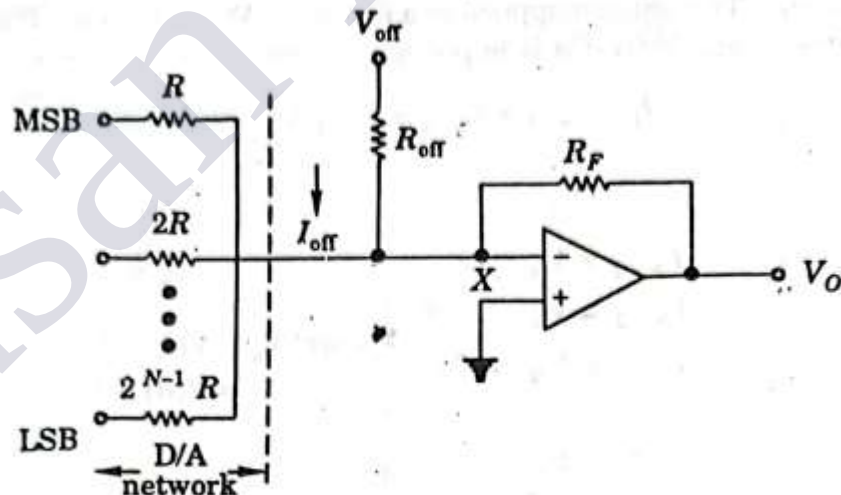
Since K=1, the formula simplifies to:

$$V_{out} = b_2 \times 4 + b_1 \times 2 + b_0$$

Now, let's calculate the analog output voltage for all possible input combinations:

| $b_2$ | $b_1$ | $b_0$ | $V_{out}$ |
|-------|-------|-------|-----------|
| 0     | 0     | 0     | 0         |
| 0     | 0     | 1     | 1         |
| 0     | 1     | 0     | 2         |
| 0     | 1     | 1     | 3         |
| 1     | 0     | 0     | 4         |
| 1     | 0     | 1     | 5         |
| 1     | 1     | 0     | 6         |
| 1     | 1     | 1     | 7         |

- ✪ **How offset voltage is used in D/A converter?**



**Fig. 10.2** Circuit used to offset the output voltage.

⊛ Example - 10.2:

- a) Design a 4 bit unipolar D/A converter with  $V(1)=-1$ ,  $V(0)=0V$  and  $R_F = 8R$ . Find the analog output voltage of the D/A converter for each of the digital inputs from 0000 to 1111.

In the general formula(10.6) we put  $N=4$  and find:

$$V_{out} = (b_3 \times 2^3 + b_2 \times 2^2 + b_1 \times 2^1 + 0 \times 2^0)$$

At first put the combination 0000 in equation and the result is 0. Similarly, we can find for every 16 combination like the following table:

| Digital input |       |       |       | Analog output |
|---------------|-------|-------|-------|---------------|
| $b_3$         | $b_2$ | $b_1$ | $b_0$ | V             |
| 0             | 0     | 0     | 0     | 0             |
| 0             | 0     | 0     | 1     | 1             |
| 0             | 0     | 1     | 0     | 2             |
| 0             | 0     | 1     | 1     | 3             |
| 0             | 1     | 0     | 0     | 4             |
| 0             | 1     | 0     | 1     | 5             |
| 0             | 1     | 1     | 0     | 6             |
| 0             | 1     | 1     | 1     | 7             |
| 1             | 0     | 0     | 0     | 8             |
| 1             | 0     | 0     | 1     | 9             |
| 1             | 0     | 1     | 0     | 10            |
| 1             | 0     | 1     | 1     | 11            |
| 1             | 1     | 0     | 0     | 12            |
| 1             | 1     | 0     | 1     | 13            |
| 1             | 1     | 1     | 0     | 14            |
| 1             | 1     | 1     | 1     | 15            |

- b) Draw the circuit of D/A converter and adjust the offset voltage for the circuit for a digital input of 1000.

Without offset, the output voltage  $V_0$  is 8 V for the digital input 1000. Therefore, the offset must produce a voltage of -8 V at the output, i.e.

$$\frac{-R_f}{R_{off}} V_{off} = -8 \text{ V}$$

We can use  $R_{off} = R$  and  $V_{off} = 1V$ . The analog voltages are given the below table:

| Digital input |       |       |       | Analog output |
|---------------|-------|-------|-------|---------------|
| $b_3$         | $b_2$ | $b_1$ | $b_0$ | V             |
| 0             | 0     | 0     | 0     | -8            |
| 0             | 0     | 0     | 1     | -7            |
| 0             | 0     | 1     | 0     | -6            |
| 0             | 0     | 1     | 1     | -5            |
| 0             | 1     | 0     | 0     | -4            |
| 0             | 1     | 0     | 1     | -3            |
| 0             | 1     | 1     | 0     | -2            |
| 0             | 1     | 1     | 1     | -1            |
| 1             | 0     | 0     | 0     | 0             |
| 1             | 0     | 0     | 1     | +1            |
| 1             | 0     | 1     | 0     | +2            |
| 1             | 0     | 1     | 1     | +3            |
| 1             | 1     | 0     | 0     | +4            |
| 1             | 1     | 0     | 1     | +5            |
| 1             | 1     | 1     | 0     | +6            |
| 1             | 1     | 1     | 1     | +7            |

- c) With the offset used above, obtain the analog output voltages if the MSB is complemented before applying to the D/A converter.

If the digital input to the D/A converter is  $\overline{b_3} b_2 b_1 b_0$ , then the analog output voltages can be obtained from the table(b) and it is given in the table below:

| Digital input |       |       |       | Analog output |
|---------------|-------|-------|-------|---------------|
| $b_3$         | $b_2$ | $b_1$ | $b_0$ | V             |
| 1             | 0     | 0     | 0     | -8            |
| 1             | 0     | 0     | 1     | -7            |
| 1             | 0     | 1     | 0     | -6            |
| 1             | 0     | 1     | 1     | -5            |
| 1             | 1     | 0     | 0     | -4            |
| 1             | 1     | 0     | 1     | -3            |
| 1             | 1     | 1     | 0     | -2            |
| 1             | 1     | 1     | 1     | -1            |
| 0             | 0     | 0     | 0     | 0             |
| 0             | 0     | 0     | 1     | +1            |
| 0             | 0     | 1     | 0     | +2            |
| 0             | 0     | 1     | 1     | +3            |
| 0             | 1     | 0     | 0     | +4            |
| 0             | 1     | 0     | 1     | +5            |
| 0             | 1     | 1     | 0     | +6            |
| 0             | 1     | 1     | 1     | +7            |

- ★ **R-2R ladder:** To analyze this circuit, for simplicity we consider a 3-bit R-2R ladder D/A network shown in Fig. 10.4. In this circuit, we have assumed the digital input as 001. The circuit is simplified using Thevenin's theorem. Applying Thevenin's theorem at XX', we obtain the circuit of Fig. 10.5a. Similarly, applying Thevenin's theorem at YY' and ZZ', we obtain the circuits of Fig. 10.5b and c, respectively. Here, LSB has been assumed as 1 and the equivalent voltage obtained is  $V/2$ .

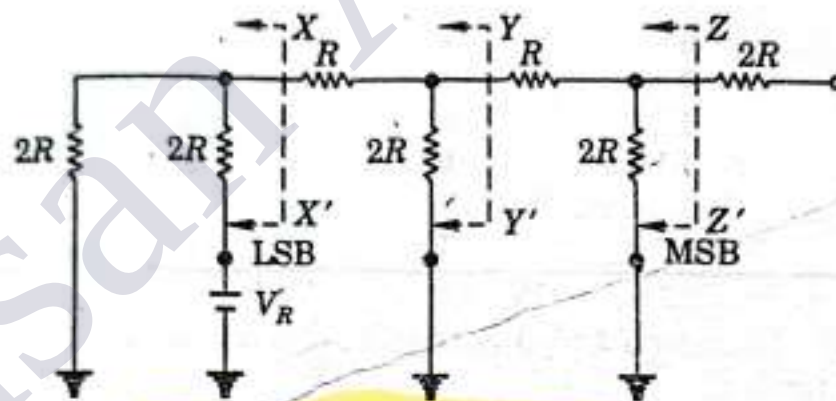


Fig. 10.4 3-bit R-2R ladder D/A network.

Similarly, for the digital input of 010 and 100, the equivalent voltages are  $V_R/2^2$  and  $V_R/2^1$ , respectively. The value of the equivalent resistance is  $3R$  in each case. Therefore, we obtain an equivalent circuit of Fig. 10.4 which is given in Fig. 10.5d. For the circuit of Fig. 10.5d, the output analog voltage  $V_O$  is given by

$$V_O = -\left(\frac{R_F}{3R} \cdot \frac{V_R}{2^3} b_0 + \frac{R_F}{3R} \cdot \frac{V_R}{2^2} b_1 + \frac{R_F}{3R} \cdot \frac{V_R}{2^1} b_2\right)$$

$$= -\left(\frac{R_F}{3R}\right) \cdot \left(\frac{V_R}{2^3}\right) [4b_2 + 2b_1 + 1b_0] \quad (10.7)$$

Equation (10.7) shows that the analog output voltage is proportional to the digital input. In general, for an  $N$ -bit D/A converter, the output voltage can be similarly determined and is given by

$$V_O = (2^{N-1} b_{N-1} + 2^{N-2} b_{N-2} + \dots + 2^2 b_2 + 2^1 b_1 + 2^0 b_0) \quad (10.8)$$

where  $R_F = 3R$  and  $V_R = -2^N V$ .

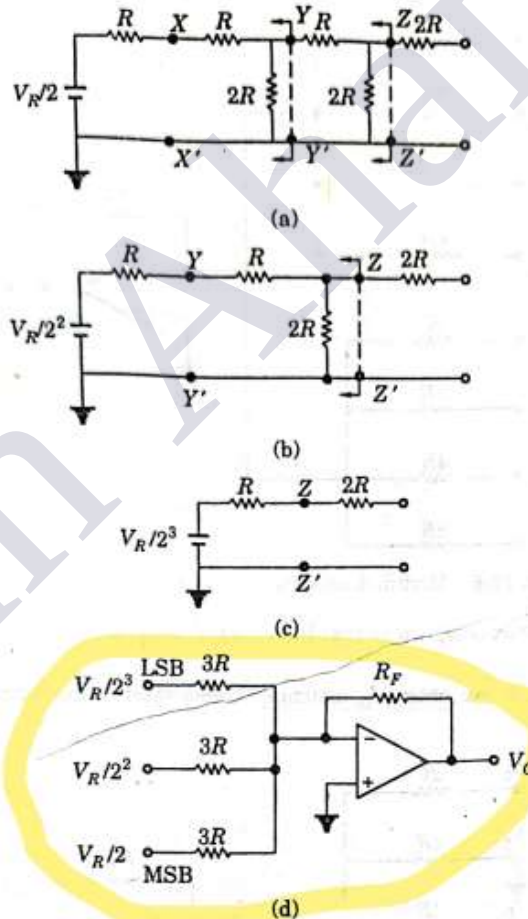


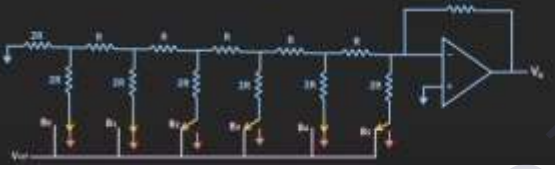
Fig. 10.5 Simplification of circuit of Fig. 10.4.

🌟 **Summary:** You can also watch the video by clicking the below ss.

**B2 B1 B0**

|       |   |                                                                                             |
|-------|---|---------------------------------------------------------------------------------------------|
| 1 0 0 | → | $\frac{V_{ref}}{2}$                                                                         |
| 0 1 0 | → | $\frac{V_{ref}}{4}$                                                                         |
| 0 0 1 | → | $\frac{V_{ref}}{8}$                                                                         |
| 1 1 1 | → | $\frac{V_{ref}}{8} + \frac{V_{ref}}{4} + \frac{V_{ref}}{2} \rightarrow \frac{7 V_{ref}}{8}$ |

For the given DAC find the full scale output voltage if  $R_f = 2 \text{ k}\Omega$  and  $R = 1 \text{ k}\Omega$ . Also find the output voltage when the input is 101100. Assume  $V_{ref} = 5V$



**101100**

$$FSO = 5V \left( \frac{-2}{1} \right) \left( \frac{1}{2^6} + \frac{1}{2^5} + \frac{1}{2^4} + \frac{1}{2^3} + \frac{1}{2^2} + \frac{1}{2^1} \right)$$

$$= -10 \times \frac{63}{64}$$

$$= -9.84375V$$



## Digital – to – Analog converter. (DAC)

### ⊛ What is quantization error? How it is reduced in A/D converter?

Quantization error happens when an analog signal is converted into a digital signal in an Analog-to-Digital (A/D) converter. It occurs because the A/D converter can only represent the signal in discrete steps, so small differences between the actual signal and its digital version are lost. This difference between the actual analog value and the closest digital value is the quantization error.

#### Example:

Imagine you are measuring temperature with a thermometer that can only show whole numbers (e.g., 25°C, 26°C). If the actual temperature is 25.5°C, you would round it to either 25°C or 26°C. The difference between the actual 25.5°C and the rounded number is the quantization error.

#### How to reduce quantization error:

1. **Increase resolution:** By using more bits in the A/D converter, the number of steps increases, making each step smaller. This reduces the gap between the actual signal and the digital value.

For example, a 3-bit converter can represent 8 levels, while a 4-bit converter can represent 16 levels. More levels mean smaller errors.

2. **Signal conditioning:** Amplifying or filtering the signal before conversion helps by ensuring that the signal is well-matched to the converter's input range, reducing errors.

**Must use graphical representation when write the answer.**

- ⊛ **Parallel comparator A/D converter:** In a resistor voltage divider network, if you have  $N$  resistors of equal value  $R$ , the total resistance between  $V_{\text{ref}}$  and ground is  $NR$ , where  $N$  is the number of resistors.

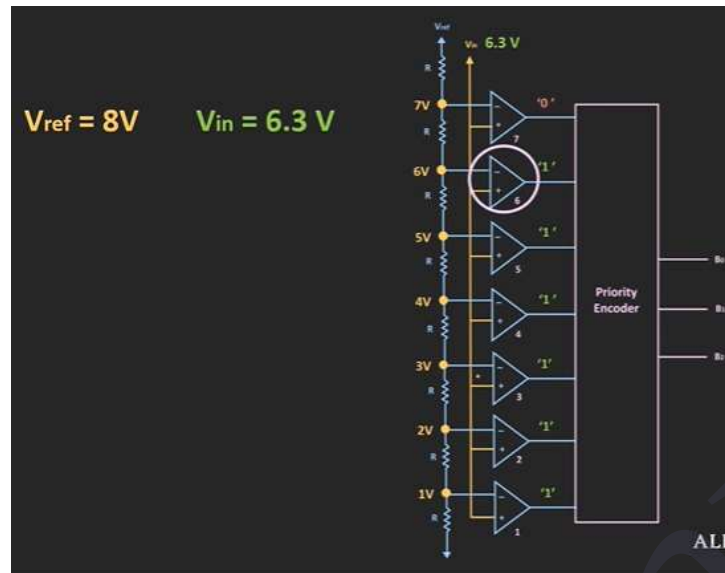
[Click here and look over the video and clear the concept.](#)

For the voltage at the  $n$ -th point (with  $n$  resistors below it):

$$V_n = V_{\text{ref}} \times \frac{nR}{NR}$$

This simplifies to:

$$V_n = V_{\text{ref}} \times \frac{n}{N}$$



★ Give the examples where D/A and A/D converters are used.

Digital-to-Analog (D/A) and Analog-to-Digital (A/D) converters are used in many everyday devices. Here are some examples:

- **D/A Converter:** Used in **audio systems** to convert digital music (like from a smartphone) into analog signals that speakers can play as sound.
- **A/D Converter:** Used in **microphones** to convert analog sound waves into digital signals for recording or processing on a computer.